SQL

Hotel Reservation System

Project Presentation

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Introduction

The Hotel Management System is a database management project developed using SQL to simplify and organize hotel operations. This project helps in managing customer details, room bookings, payments, and staff information efficiently through a structured database system.

The system uses SQL tables, keys, relationships, and queries to store and retrieve data accurately. By using primary keys and foreign keys, different tables are connected to maintain data consistency and reduce redundancy.

This project demonstrates the practical use of SQL commands and functions in real-world applications and improves the efficiency, accuracy, and security of hotel management operations.

Project Objectives

1

To manage hotel booking and customer records efficiently using SQL.

2

To create relationships between tables using primary key and foreign key.

3

To perform data operations and generate reports using SQL queries and functions.

ER Diagram (Entity Relationship Diagram)

The ER Diagram represents the relationship between different tables used in the Hotel Reservation System.

Relationship Explanation

- One customer can make many bookings.
- One room can be booked many times.
- Each booking is connected to one payment.
- Staff table is independent and stores hotel employee details.

Key

PK = Primary Key

FK = Foreign Key

SQL Concepts Used

DDL Commands (CREATE, ALTER, DROP)
DML Commands (INSERT, UPDATE, DELETE)
DQL Commands (SELECT Queries)
Constraints (PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE)
Joins (INNER JOIN, LEFT JOIN, RIGHT JOIN)
Aggregate Functions

Hotel Reservation System Implementation


```sql
-- create customer table
CREATE TABLE customer (
  customer_id INT PRIMARY KEY,
  customer_name VARCHAR(50),
  gender VARCHAR(10),
  phone VARCHAR(15),
  city VARCHAR(30)
);
```

CUSTOMER
- customer_id (PK)
- customer_name
- gender
- phone
- city

1 : M

BOOKING
- booking_id (PK)
- customer_id (FK)
- room_id (FK)
- check_in
- check_out
- total_days

M : 1

ROOM
- room_id (PK)
- room_type
- price_per_day
- availability

PAYMENT
- payment_id (PK)
- booking_id (FK)
- amount
- payment_mode
- payment_status

```sql
-- create room table
CREATE TABLE room (
  room_id INT PRIMARY KEY,
  room_type VARCHAR(30),
  price_per_day DECIMAL(10,2),
  availability VARCHAR(10)
);
```

1 : 1

customer_id	customer_name	gender	phone	city
1	Rohit Sharma	Male	9876543210	Delhi
2	Priya Verma	Female	9123456780	Mumbai
3	Amit Singh	Male	9988776655	Jaipur

booking_id	customer_id	room_id	check_in	check_out	total_days
201	1	101	2024-05-01	2024-05-04	3
202	2	102	2024-05-02	2024-05-05	3
203	3	103	2024-05-03	2024-05-06	3

room_id	room_type	price_per_day	availability
101	single	1500.00	yes
102	double	2500.00	yes
103	suite	5000.00	no
104	single	1800.00	yes

```sql
-- create payment table
CREATE TABLE payment (
  payment_id INT PRIMARY KEY,
  booking_id INT,
  amount DECIMAL(10,2),
  payment_mode VARCHAR(20),
  payment_status VARCHAR(20),
  FOREIGN KEY (booking_id)
  REFERENCES booking(booking_id)
);
```

1 : 1

payment_id	booking_id	amount	payment_mode	payment_status
301	201	4500.00	card	paid
302	202	7500.00	upi	paid
303	203	15000.00	cash	paid

LEGEND
- PK : Primary Key
- FK : Foreign Key
- 1 : M : One to Many
- M : 1 : Many to One
- 1 : 1 : One to One
- ------- Relationship

```sql
-- create staff table
CREATE TABLE staff (
  staff_id INT PRIMARY KEY,
  staff_name VARCHAR(50),
  department VARCHAR(30),
  salary DECIMAL(10,2)
);
```

STAFF
- staff_id (PK)
- staff_name
- department
- salary

staff_id	staff_name	department	salary
1	Neha Gupta	Reception	25000.00
2	Rajesh Kumar	Housekeeping	22000.00
3	Suresh Yadav	Management	40000.00

KEY FEATURES
- One customer can make many bookings.
- One room can be booked many times.
- Each booking has one payment.
- Staff table stores employee details.

Hotel Reservation System Implementation

-- 1. Database Creation and Use Query

Create database hotel_reservation

```
create database hotel_reservation;
use hotel_reservation;
```

Hotel Reservation System Implementation

```sql
insert into customer values
(1, 'rahul', 'male', '9876543210', 'surat'),
(2, 'priya', 'female', '9876501234', 'mumbai'),
(3, 'amit', 'male', '9876512345', 'delhi'),
(4, 'neha', 'female', '9988776655', 'ahmedabad'),
(5, 'karan', 'male', '9090909090', 'pune'),
(6, 'riya', 'female', '8080808080', 'jaipur'),
(7, 'vijay', 'male', '7070707070', 'rajkot'),
(8, 'pooja', 'female', '6060606060', 'baroda');
```

-- create a customer table

customer_id	customer_name	gender	phone	city
1	rahul	male	9876543210	surat
2	priya	female	9876501234	mumbai
3	amit	male	9876512345	delhi
4	neha	female	9988776655	ahmedabad
5	karan	male	9090909090	pune
6	riya	female	8080808080	jaipur
7	vijay	male	7070707070	rajkot
8	pooja	female	6060606060	baroda
NULL	NULL	NULL	NULL	NULL

Hotel Reservation System Implementation

```
-- room table
create table room (
    room_id int primary key,
    room_type varchar(30),
    price_per_day decimal(10,2),
    availability varchar(10)
);
```

-- create a room table

room_id	room_type	price_per_day	availability
101	single	1500.00	yes
102	double	2500.00	yes
103	suite	5000.00	no
104	single	1800.00	yes
105	double	2700.00	no
106	suite	6000.00	yes
NULL	NULL	NULL	NULL

Hotel Reservation System Implementation

```
-- booking table with foreign keys
create table booking (
    booking_id int primary key,
    customer_id int,
    room_id int,
    check_in date,
    check_out date,
    total_days int,
constraint fk_customer
foreign key (customer_id)
references customer(customer_id),
constraint fk_room
foreign key (room_id)
references room(room_id)
);
```

-- create a booking table

booking_id	customer_id	room_id	check_in	check_out	total_days
1001	1	101	2026-05-01	2026-05-03	2
1002	2	102	2026-05-05	2026-05-08	3
1003	3	103	2026-05-10	2026-05-12	2
1004	4	104	2026-05-15	2026-05-18	3
1005	5	105	2026-05-20	2026-05-23	3
1006	6	106	2026-05-25	2026-05-27	2
NULL	NULL	NULL	NULL	NULL	NULL

Hotel Reservation System Implementation

```sql
-- payment table with foreign key
create table payment (
    payment_id int primary key,
    booking_id int,
    amount decimal(10,2),
    payment_mode varchar(20),
    payment_status varchar(20),

    constraint fk_booking
    foreign key (booking_id)
    references booking(booking_id)
);
```

-- create a payment table

payment_id	booking_id	amount	payment_mode	payment_status
201	1001	3000.00	cash	paid
202	1002	7500.00	upi	paid
203	1003	10000.00	card	pending
204	1004	5400.00	cash	paid
205	1005	8100.00	upi	pending
206	1006	12000.00	card	paid
NULL	NULL	NULL	NULL	NULL

Hotel Reservation System Implementation

```
-- staff table
create table staff (
    staff_id int primary key,
    staff_name varchar(50),
    department varchar(30),
    salary decimal(10,2)
);
```

-- create a staff table

staff_id	staff_name	department	salary
1	mahesh	manager	50000.00
2	sneha	reception	25000.00
3	rakesh	cleaning	18000.00
4	anita	security	22000.00
5	vishal	maintenance	30000.00
NULL	NULL	NULL	NULL

Hotel Reservation System Implementation

```
-- available rooms
select * from room
where availability = 'yes';
```

--Display a list of all rooms that are currently available for booking.

Result Grid | Filter Rows: | Edit:

room_id	room_type	price_per_day	availability
101	single	1500.00	yes
102	double	2500.00	yes
104	single	1800.00	yes
106	suite	6000.00	yes
NULL	NULL	NULL	NULL

Hotel Reservation System Implementation

```sql
-- total paid payment
select sum(amount) as total_payment
from payment
where payment_status = 'paid';
```

--Calculate the total revenue generated from all successfully completed ('paid') payments.

Result Grid | Filter Rows:

total_payment
27900.00

Hotel Reservation System Implementation

```
-- total rooms
select count(*) as total_rooms
from room;
```

--Find the total number of rooms managed by the hotel/property.

Result Grid | Filter Rows

total_rooms
6

Hotel Reservation System Implementation

```
-- customers from surat
select * from customer
where city = 'surat';
```

--Retrieve the complete details of all customers who reside in the city of Surat.

	customer_id	customer_name	gender	phone	city
▶	1	rahul	male	9876543210	surat
*	NULL	NULL	NULL	NULL	NULL

Thank You

We appreciate your time and interest in our project.